04-25 第一篇

Undersea Volcanoes: Hidden Giants in the Ocean

Most people know about volcanoes on land, but did you know that most volcanoes on Earth are actually under the sea? These undersea volcanoes, also called submarine（水下的） volcanoes, form along the ocean floor. [1] In fact, scientists believe there are more than one million undersea volcanoes around the world. Many of them are still active, releasing heat and gases into the ocean. Some rise high enough to break the surface and become islands—the Hawaiian Islands were formed this way millions of years ago.

How do undersea volcanoes form? Deep under the ocean, the Earth's crust（地壳） is thinner than on land. [2] When hot magma（岩浆） from inside the Earth pushes up through weak spots, it meets cold seawater. The magma cools quickly and hardens into new rock. Over time, layer upon layer of lava builds up, creating mountains under the water. This process has been going on for billions of years and has shaped the ocean floor we see today.

Undersea volcanoes do more than build mountains. They also support unique forms of life. [3] In the dark, hot waters around volcanic vents（火山喷口）, special bacteria and other creatures have learned to live without sunlight. They get energy from chemicals in the hot water instead. Scientists have found strange fish, crabs, and tube worms living in these extreme conditions. Such discoveries have changed our understanding of where life can exist.

Studying undersea volcanoes helps scientists understand not only the ocean but also the whole planet. [4] By mapping the seafloor and monitoring volcanic activity, researchers can learn how the Earth's plates move and where earthquakes might occur. Some countries have set up stations to watch undersea volcanoes, hoping to give early warning of possible tsunamis（海啸）. This knowledge can help keep coastal communities safe.

[5] From shaping islands to supporting strange forms of life, undersea volcanoes remind us that our planet is still full of surprises. Much of the ocean remains unexplored, and new volcanic features are being discovered every year. As technology improves, we will surely learn even more about these hidden giants beneath the waves.

选项

A. In short, undersea volcanoes play a vital role in both the geology and the biology of our oceans. B. In other words, these volcanoes are much more common than the ones we see on land. C. The way they form is linked to the movement of the Earth's plates and the heat inside our planet. D. That is why scientists from different fields work

together to study these underwater wonders. E. These hot spots create a home for organisms that cannot survive anywhere else. F. Some of these volcanoes are so large that they would be among the tallest mountains on land.

答案键（Answer Key）

[1] B [2] C [3] E [4] D [5] A

深度解析

- **[1] B**：
  - **线索**：引出海底火山的数量与普遍性。空前描述"most volcanoes on Earth are actually under the sea...form along the ocean floor"，空后具体说明"In fact, scientists believe there are more than one million undersea volcanoes...Many of them are still active...Some rise high enough to become islands"。B选项"In other words, these volcanoes are much more common than the ones we see on land"（换言之，这些火山比我们在陆地上看到的要多得多）中"In other words"承接空前"most volcanoes...under the sea"，"much more common"与空后"more than one million"形成呼应，衔接顺畅。
- **[2] C**：
  - **线索**：海底火山形成机制的总领。空前说明"How do undersea volcanoes form? Deep under the ocean, the Earth's crust is thinner"，空后详述"When hot magma pushes up through weak spots...meets cold seawater...cools and hardens...layer upon layer builds up...This process has been going on for billions of years"。C选项"The way they form is linked to the movement of the Earth's plates and the heat inside our planet"（它们的形成方式与地球板块运动和地球内部热量有关）中"the way they form"与空前问句对应，"Earth's plates...heat inside"与空后"magma from inside the Earth...weak spots"逻辑衔接，为后文具体描述做铺垫。
- **[3] E**：
  - **线索**：海底火山附近独特生物环境的概括。空前描述"They also support unique forms of life...In the dark, hot waters around volcanic vents, special bacteria and other creatures have learned to live without sunlight"，空后列举"They get energy from chemicals...found strange fish, crabs, tube worms...extreme conditions...changed our understanding of where life can exist"。E选项"These hot spots create a home for organisms that cannot survive anywhere else"（这些热点为在其他地方无法生存的生物创造了家园）中"These hot spots"指代火山喷口周围区域，"organisms that cannot survive anywhere else"与空后"live without sunlight...extreme conditions"完美对应，衔接自然。

- [4] D：
  - **线索**：研究海底火山的科学意义与跨领域合作。空前描述"Studying undersea volcanoes helps...understand not only the ocean but also the whole planet...By mapping the seafloor and monitoring volcanic activity...learn how the Earth's plates move...Some countries have set up stations...early warning of tsunamis"，空后说明"This knowledge can help keep coastal communities safe"。D选项"That is why scientists from different fields work together to study these underwater wonders"（因此来自不同领域的科学家共同研究这些水下奇观）中"That is why"承接空前研究的多重价值，"different fields work together"与空前"mapping...monitoring...plates move...early warning"形成呼应，说明研究需要多学科协作，衔接精准。
- [5] A：
  - **线索**：全文收束与升华。空[5]位于结尾段段首，空后列举"From shaping islands to supporting strange forms of life...planet is still full of surprises...new volcanic features...learn even more about these hidden giants"。A选项"In short, undersea volcanoes play a vital role in both the geology and the biology of our oceans"（简言之，海底火山在海洋的地质和生物两方面都发挥着重要作用）用"In short"总结全文，"geology and biology"与空后"shaping islands...supporting strange forms of life"完美呼应，收束得当。
- **干扰项分析**：
  - **F. Some of these volcanoes are so large that they would be among the tallest mountains on land.**：强调海底火山体型巨大，易被误填[1]或[2]。但F为具体事实描述，与空[1]前后"more than one million...still active...form islands"的数量与活动性主题不完全匹配，B的"much more common"更贴合；空[2]需形成机制的总领句，F讲的是火山规模，与形成过程无关。F属"主题相关但逻辑功能错位"型干扰。